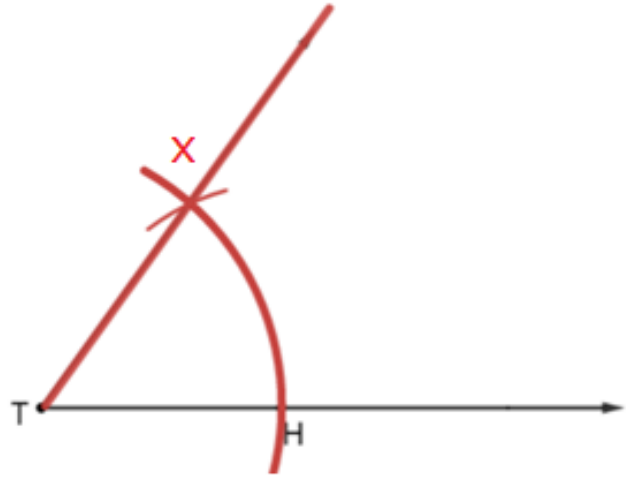
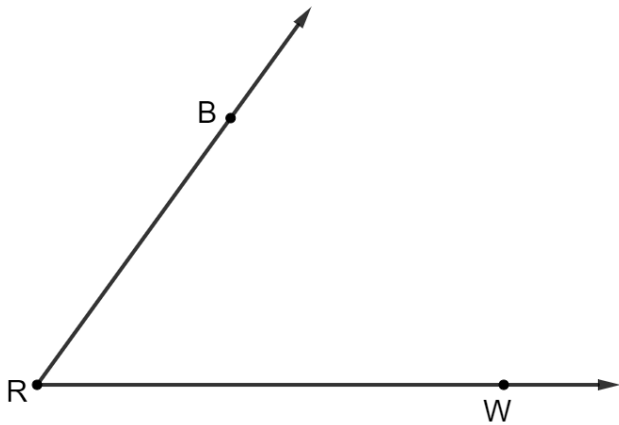


Geometry
Unit 1
Lesson 9 Practice

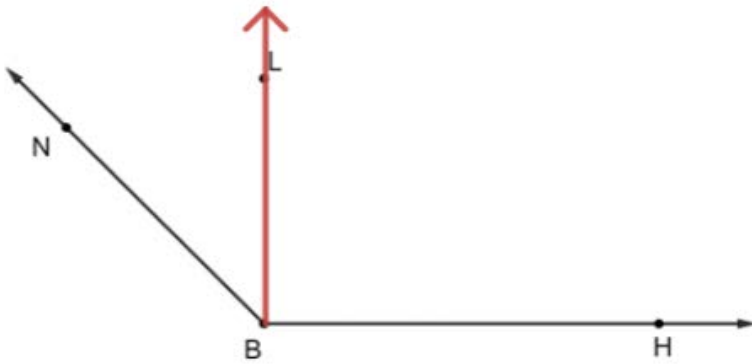
Name Answer Key
Date _____
Period _____

Angle BRW and ray TH are shown.



1. Use $\overrightarrow{TH}$ to create $\angle HTX$, a copy of $\angle BRW$.

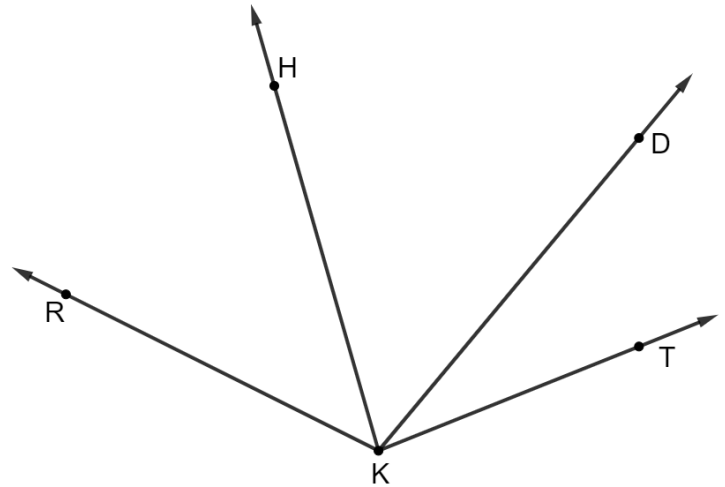
Point L lies in the interior of angle NBH as shown.



2. Draw ray BL .
3. Complete the statement.

Since point L is in the interior of $\angle NBH$ then $m\angle NBL + m\angle LBH = m\angle NBH$.

Angles RKH , HKD , and DKT are shown where $m\angle RKH = (4y)^\circ$, $m\angle DKH = (6y - 16)^\circ$, $m\angle DKT = (2y + 4)^\circ$, and $m\angle RKT = 132^\circ$.



4. Determine the value of y .
 $4y + 6y - 16 + 2y + 4 = 132$
 $12y - 12 = 132$
 $12y = 144$
 $y = 12$

5. Determine $m\angle HKT$.
 $6y - 16 + 2y + 4$
 $8y - 12$
 $8 \cdot 12 - 12$
 84°

6. Determine $m\angle DKR$.
 $6y - 16 + 4y$
 $10y - 16$
 $10 \cdot 12 - 16$
 104°

Points R , H , M , and D lie in the interior of straight angle YWL such that $m\angle YWM = 105^\circ$, $m\angle MWD = 43^\circ$, $m\angle HWD = 59^\circ$, and $m\angle RWH = 2m\angle HWM$.

Determine each.

7. $m\angle YWD$
 $105^\circ + 43^\circ = 148^\circ$
8. $m\angle LWD$
 $180^\circ - 148^\circ = 32^\circ$
9. $m\angle HWM$
 $59^\circ - 43^\circ = 16^\circ$
10. $m\angle HWR$
 $2 \cdot 16^\circ = 32^\circ$

